

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch: Mechanical Engineering Semester: V Subject Code & Name: BTMC506 & Applied Thermodynamics Max Marks: 60 Date: 22-08-2023 Duration: 3 Hr			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the requirements of good fuel	BL₁	6
B)	What is meant by term fuel? Classify fuels in detail	BL₂	6
C)	The volumetric analysis of a gas is given below as, CO ₂ = 14%, CO = 1%, O ₂ = 5%, N ₂ = 80% Obtain the percentage analysis by mass.	BL₄	6
Q.2	Solve Any Two of the following.		12
A)	Describe with neat diagram, the construction and working of a low-pressure boiler with its specifications	BL₄	6
B)	Compare the water tube boiler with fire tube boiler. Also explain the boiler mountings and accessories.	BL₃	6
C)	Explain the Indian Boiler Regulation Act	BL₃	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the Rankine vapour cycle in detail with P-V and T-S graph	BL₄	6
B)	A turbine is supplied with steam at a pressure of 32 bar and a temperature of 410 °C. The steam then expands isentropically to a pressure of 0.08 bar. Find the dryness fraction at the end of expansion Note: Use mollier chart and represent it with neat sketch	BL₅	6
C)	Consider an air standard cycle in which the air enters the compressor at 1.0 bar and 20°C. The pressure of air leaving the compressor is 3.5 bar and the temperature at turbine inlet is 600°C. Determine per kg of air: (i) Efficiency of the cycle, (ii) Heat supplied to air, (iii) Work available at the shaft, (iv) Heat rejected in the cooler, and	BL₅	6

	(v) Temperature of air leaving the turbine. For air $\gamma = 1.4$ and $c_p = 1.005 \text{ kJ/kg K}$.		
Q.4	Solve Any Two of the following.		12
A)	Explain the term compounding of steam turbine. Also draw a sketch of velocity compounding and labelled it.	BL₃	6
B)	A single stage impulse turbine motor has a diameter of 1.2 m running at 3000 rpm. The nozzle angle is 18° , blade speed ratio is 0.42. The ratio of relative velocity at outlet to relative velocity at inlet is 0.9. The outlet blade angle is 30° small than inlet blade angle. The steam flow rate is 5 Kg/sec. Draw a velocity diagram.	BL₅	6
C)	Explain the terms 1. Vacuum Efficiency 2. Condenser efficiency	BL₃	6
Q. 5	Solve Any Two of the following.		12
A)	Classify air compressors. Also state the applications of compressed Air	BL₃	6
B)	Explain the working of single stage reciprocating compressor with indicator (P-V) diagram	BL₄	6
C)	A single acting single stage reciprocating compressor of bore 300 mm and stroke 400 mm is required to compress air from 1 bar to 5 bar. Find the power required to drive the compressor, if it runs at 200 r.p.m. and compression is, Isothermal Adiabatic with index = 1.4 Polytropic with index = 1.3	BL₆	6
	*** End ***		